```
import random
n = int(input("Enter number of packages: "))
weights = []
values = []
for i in range(n):
    w = int(input("Enter weight of package " + str(i + 1) + ": "))
    v = int(input("Enter value of package " + str(i + 1) + ": "))
    weights.append(w)
    values.append(v)
capacity = int(input("Enter truck capacity: "))
def fitness(chromosome):
    total_weight = 0
    total_value = 0
    for i in range(n):
        if chromosome[i] == 1:
            total_weight += weights[i]
            total_value += values[i]
    if total_weight > capacity:
        return 0
    return total_value
def create_chromosome():
    return [random.randint(0, 1) for i in range(n)]
def crossover(parent1, parent2):
    point = random.randint(1, n - 1)
    child = parent1[:point] + parent2[point:]
    return child
def mutate(chromosome):
    for i in range(n):
        if random.random() < 0.1:
            chromosome[i] = 1 - chromosome[i]
population = []
for i in range(10):
    population.append(create_chromosome())
for generation in range(50):
    population.sort(key=fitness, reverse=True)
    new_population = population[:2]
    while len(new_population) < 10:
        parent1 = random.choice(population[:5])
        parent2 = random.choice(population[:5])
        child = crossover(parent1, parent2)
        mutate(child)
        new_population.append(child)
    population = new_population
best = max(population, key=fitness)
total_weight = 0
total_value = 0
for i in range(n):
    if best[i] == 1:
        total_weight += weights[i]
        total_value += values[i]
print("\nBest chromosome:", best)
print("Selected packages:")
for i in range(n):
    if best[i] == 1:
        print("Package", i + 1)
print("Total weight:", total_weight)
print("Total value:", total_value)
```

```
Enter number of packages: 5
Enter weight of package 1: 2
Enter value of package 1: 12
Enter weight of package 2: 1
Enter value of package 2: 10
```

```
Enter weight of package 3: 3
Enter value of package 3: 20
Enter weight of package 4: 2
Enter value of package 4: 5
Enter weight of package 5: 4
Enter value of package 5: 25
Enter truck capacity: 7

Best chromosome: [1, 1, 0, 0, 1]
Selected packages:
Package 1
Package 2
Package 5
Total weight: 7
Total value: 47
```